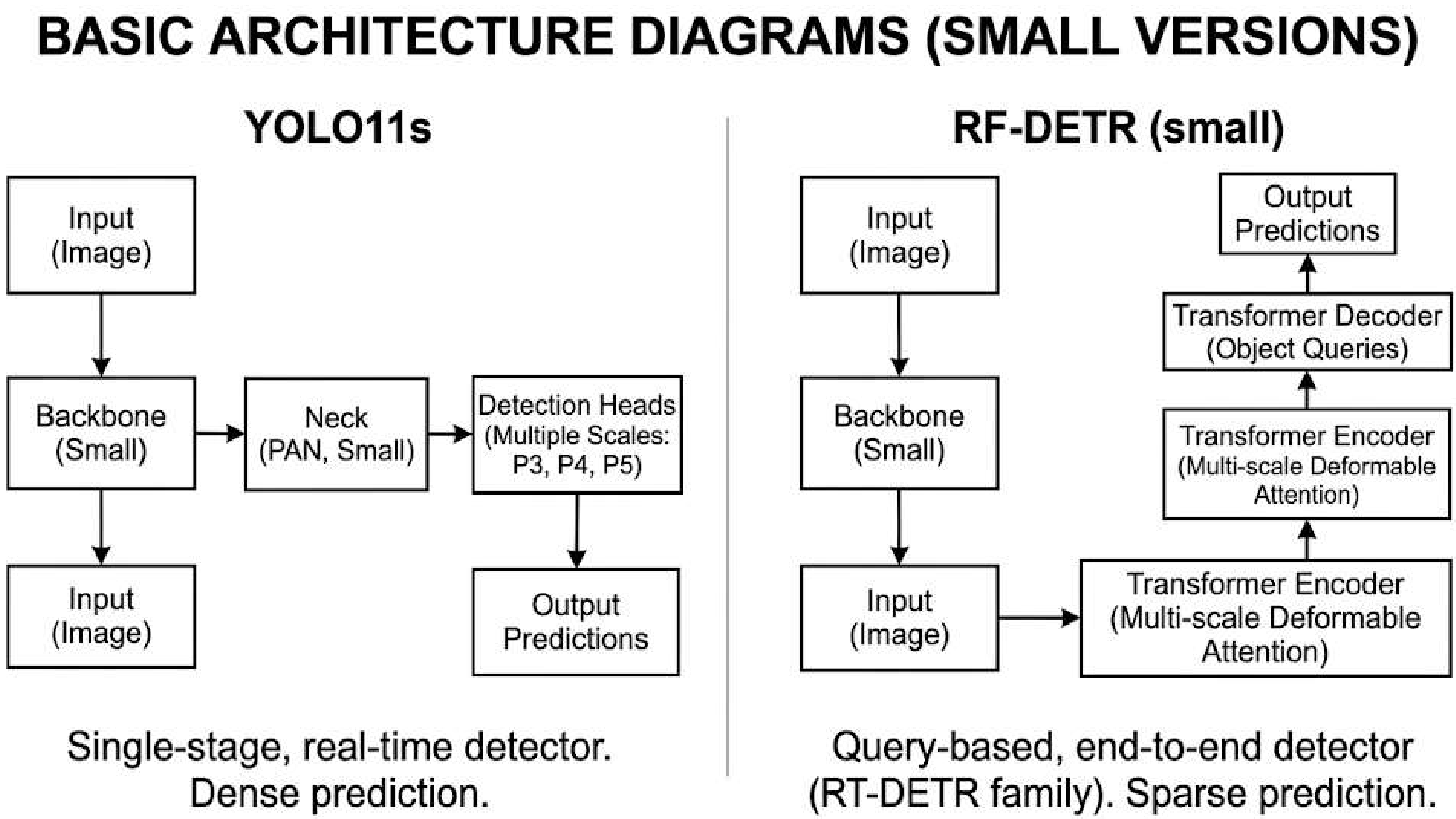
# Maritime detection:
# YOLO11s Fine tuning vs RF-DETR Full Fine Tuning

By S Sriram


BASIC ARCHITECTURE DIAGRAMS (SMALL VERSIONS)
YOLO11s
Input (Image)
Backbone (Small)
Neck (PAN, Small)
Detection Heads (Multiple Scales: P3, P4, P5)
Input (Image)
Output Predictions
Single-stage, real-time detector. Dense prediction.
RF-DETR (small)
Input (Image)
Backbone (Small)
Input (Image)
Transformer Encoder (Multi-scale Deformable Attention)
Transformer Encoder (Multi-scale Deformable Attention)
Transformer Decoder (Object Queries)
Output Predictions
Query-based, end-to-end detector (RT-DETR family). Sparse prediction.

# Fine-Tuning Strategy Used

- YOLO11s: As YOLO allows layer freezing in default during fine tuning, initial layers of YOLO were frozen as they capture broader attributes like colour, shapes etc that are common between maritime objects and real world objects.

- RF-DETRs: Since RF-DETR itself protects its weights by using a more conservative learning rate and doesn’t have a native option of freezing layers like YOLO, we’ve decided to take a full fine tuning approach.

# Overall test metrics

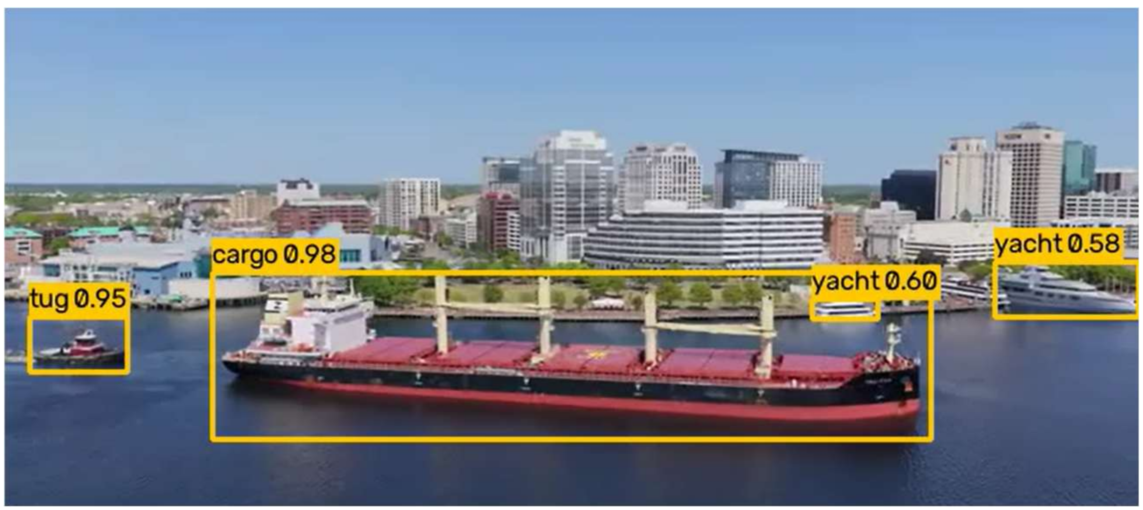
| Metric | YOLO11s (Partial Fine-Tune) | RF-DETR-Small (Full Fine-Tune) |
|---|---|---|
| **mAP@50** | 0.94 | **0.96** |
| **mAP@50-95** | 0.83 | **0.92** |
| **Recall / AR** | 0.89 (Recall) | **0.93 (Average Recall)** |

# Per class Evaluation metrics

| Class | YOLO11s(AP@50) | RF-DETR(AP@50) | YOLO11s(AP@50-95) | RF-DETR(AP@50-95) |
|---|---|---|---|---|
| **boat** | 0.883 | **0.977** | 0.737 | **0.918** |
| **buoy** | **0.942** | 0.911 | 0.852 | **0.871** |
| **cargo** | 0.960 | **0.970** | 0.879 | **0.951** |
| **ferry** | 0.835 | **0.880** | 0.713 | **0.800** |
| **tug** | 0.993 | **1.000** | 0.845 | **0.966** |
| **warship** | 0.989 | **1.000** | 0.915 | **0.983** |
| **yacht** | 0.991 | **0.995** | 0.912 | **0.967** |

**YOLO 11s**


**RF-DETR Small**




- In this case, YOLO struggles to identify the large cargo ship in the centre of the image and also misses the Yachts far away in the background.

- On the other hand RF-DETR showcases the perfect example of the difference between a attention based and a convolution based model. As RF-DETR is attention based is correctly detects the Cargo and the Tug in the centre of the image. It also detects 2 Yachts in the background.

**YOLO 11s**


**RF-DETR Small**

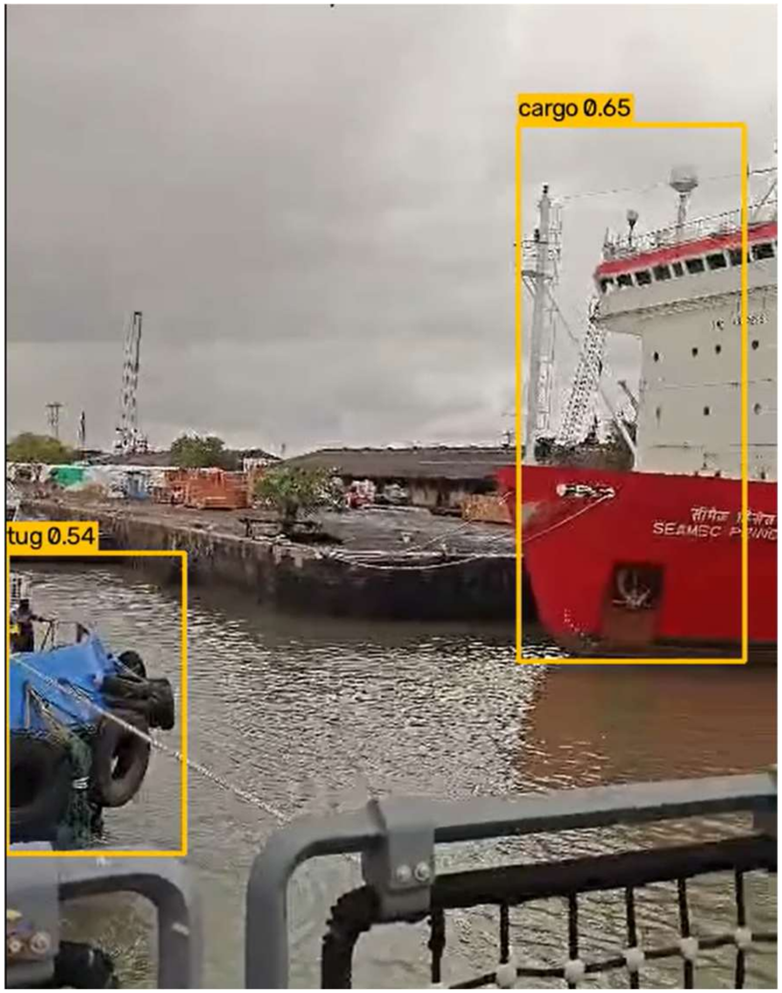

- In this case, YOLO performs much better than RF-DETR by correctly identifying the Cargo and the Yacht.
- RF-DETR identifies the Cargo as a Tug with a 92 percent confidence which is a failure. One place where the RF-DETR does well is in building a sharp and precise bounding box around the Cargo in comparison to YOLO.

**YOLO 11s**

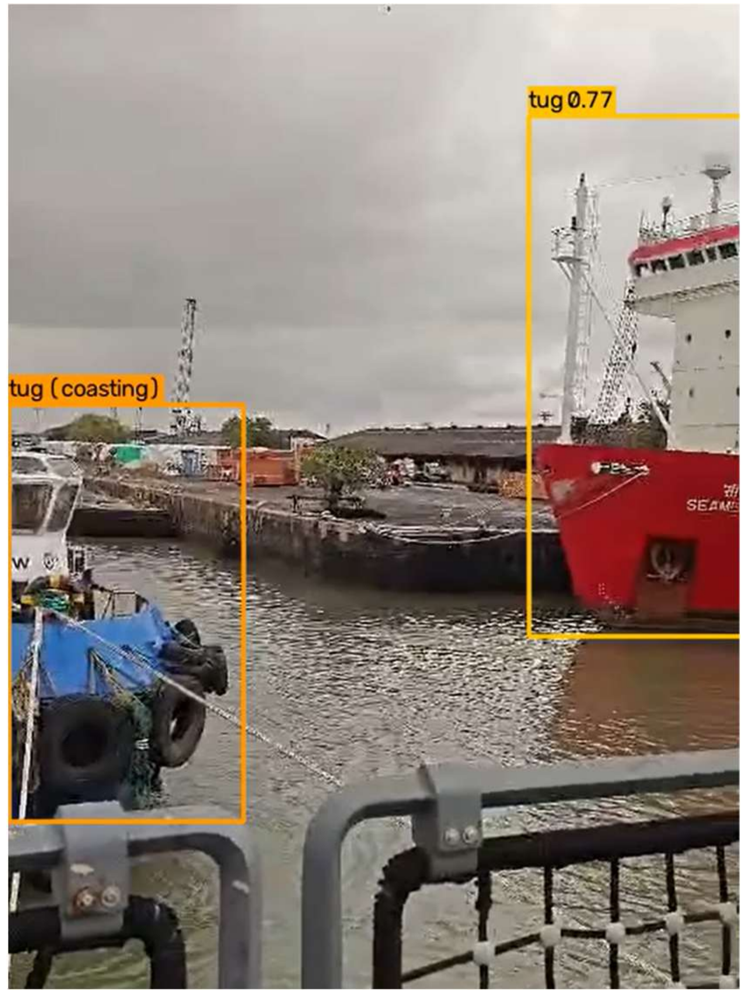

**RF-DETR Small**


- This is another demonstration of the advantage of using RF-DETR over YOLO.
- Both models do well in this case but RF-DETR detects the objects with higher confidence and also produces a tighter bounding box around them.

**YOLO 11s**




**RF-DETR Small**




- In this case, YOLO takes a little more time but classifies the Cargo correctly while the RF-DETR incorrectly classifies it as a Tug.

- Again the bounding boxes are better in the RF-DETR output.

# Key Takeaways

•**Low Training Volume:** The dataset contains a relatively low number of total instances (3,192 training instances across all categories). Transformer-based architectures typically require massive datasets to perfectly learn decision boundaries from scratch.

•**High Intra-Class Similarity:** The 7 chosen classes (boat, buoy, cargo, ferry, tug, warship, yacht) are broad. Vessels frequently share overlapping visual attributes like hull shapes, wake patterns, and bridge structures, leading to confusion when distinguishing between a large tug and a small cargo ship.

•**Architectural Trade-offs (Attention vs. Convolution):**

- **RF-DETR (Global Attention):** Excels at mapping global spatial context, allowing it to capture highly accurate bounding boxes and detect distant background objects without NMS post-processing. However, it can over-index on global shapes and miss the distinct local features required for robust classification.
- **YOLO11s (CNN-Based):** Retains strong local inductive biases due to its convolutional layers, making it more robust at extracting specific local textures and features needed to correctly classify visually similar vessels.